WEAKLY ETALE RING MAPS ARE CLOSED UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The base change of a flat ring map is flat. invoked at proof lines 30368-30369. Keeps the map C tensor_A C -> C tensor_B C flat.
- The composition of flat ring maps is flat. invoked at proof lines 30369-30371. Composes the two flat maps in the factorization and gives part (2).

Source reference. Original lemma 00HI: sources/stacks-chow/tags.txt, source lines 644–644. Cited at proof line 30368 for the flatness of the first factorization map; the conclusion is used inside the counted span.

Source reference. Original lemma 00HC: sources/stacks-chow/tags.txt, source lines 638–638. Cited at proof lines 30368-30371 for the flatness of the composition and part (2); the conclusion is used inside the counted span.

Lemma 0.2. *Let $A \to B$ and $B \to C$ be ring maps.*

(1) *If $B \otimes_A B \to B$ and $C \otimes_B C \to C$ are flat, then $C \otimes_A C \to C$ is flat.*
(2) *If $A \to B$ and $B \to C$ are weakly étale, then $A \to C$ is weakly étale.*

Proof. Part (1) follows from the factorization

$$C \otimes_A C \longrightarrow C \otimes_B C \longrightarrow C$$

of the multiplication map, the fact that

$$C \otimes_B C = (C \otimes_A C) \otimes_{B \otimes_A B} B,$$

the fact that a base change of a flat map is flat, and the fact that the composition of flat ring maps is flat. See Algebra, Lemmas 00HI and 00HC. Part (2) follows from (1) and the fact (just used) that the composition of flat ring maps is flat. □

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